

FINAL YEAR PROJECT REPORT

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Declaration

This project was carried out jointly by myself and my lab partner, Harry Tonge. Together, we collected and analysed the experimental data and produced the final results.

I was responsible for the design and CAD modelling (in Autodesk Inventor) of the 3D-printed slit modules, including the single-slit barriers, double-slit centre pieces, and the redesigned solid launcher adapted from the design of Pucci et al. [1]. Together, Harry and I wrote and developed the Python/OpenCV automated tracking pipeline (`Tracking_Algorithm.py`), and the single-slit trajectory analysis code (`Trajectory_Reader_Single_slit.py`), with the assistance of ChatGPT 5.4. Harry wrote the double-slit trajectory reader (`Trajectory_Reader_Double_slit.py`), based on the single-slit trajectory reader. We jointly modified the Arduino code controlling the relay timing for the droplet generation cycle; Harry further developed the two-Arduino system for fully automated operation during Week 17.

Together, Harry and I calibrated the laser displacement detector, performed data collection for the single-slit experiments and jointly set up and tested the double-slit experiment. Harry collected the resulting data for the double-slit experiment in my absence in Week 17. The interpretation of the overall experimental programme was carried out jointly. Our supervisor, Dr Massimo Antognozzi, helped guide the interpretation of our results, including suggesting the comparison with de Broglie and Bohmian frameworks. Lab technician, Tom Kennedy, provided practical assistance throughout, including construction of the displacement detector, assistance with the automated bath on/off generation system, and advice on hardware modifications. The apparatus was originally developed by Grace Casson, Ben Machin and Tom Kennedy in the previous academic year [2].

All data analysis, figure production, and interpretation presented in this report are my own, except where otherwise indicated: the theoretical diffraction maps in Figure 12 are reproduced from Darrow and Bush [3], and the slit cross-section diagrams (Figures 3 and 4) are adapted from Pucci et al. [1]. See below for AI usage.

AI Declaration

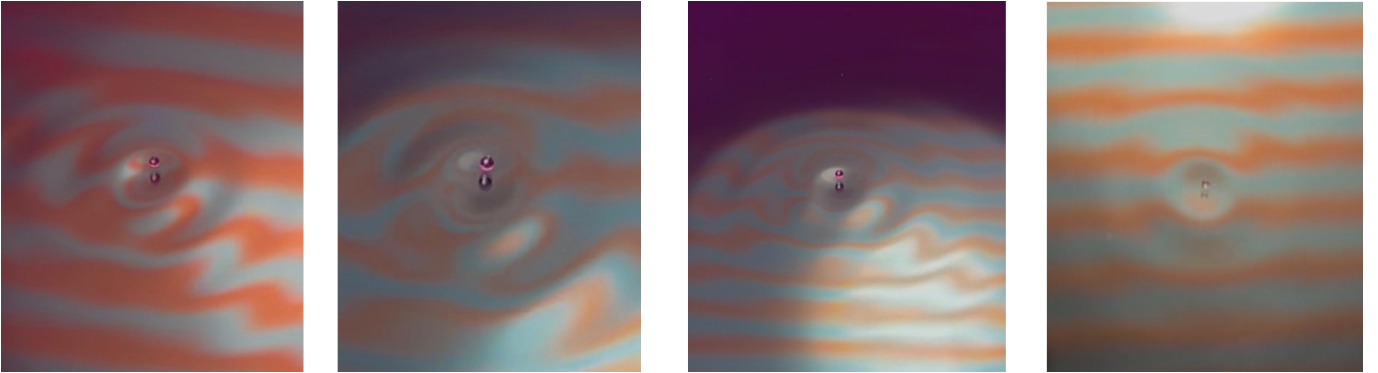
AI tools (ChatGPT 5.4) were used to assist with the development and debugging of the Python code for the camera interface and automated droplet tracking system. ChatGPT 5.4 was also used to help troubleshoot experimental problems during the project. Claude (Opus 4.6) was used for report planning, structural feedback, and checking physics arguments during the drafting process. All AI-assisted work is documented in the accompanying lab notebook. No AI tools were used to generate experimental results, analyse data, or produce figures. The analysis, interpretation, and writing of this report are entirely my own.

Acknowledgements

I would like to thank my supervisor, Dr Massimo Antognozzi, for his guidance and enthusiasm throughout the year, and for shaping the experimental direction of this project. Thanks are also due to my lab partner, Harry Tonge, for his contributions to data collection, code development, and many useful discussions. I am also grateful to lab technician Tom Kennedy for his practical help with the numerous hardware modifications this experiment required. Thanks to Freddy Millar and Lauren Matthews for useful discussions, and to Dr Karim Thébault for reviewing our results and the interesting conversations that followed. Finally, I would like to acknowledge Grace Casson and Ben Machin, whose work in the previous academic year established the apparatus on which this project builds.

Abstract

A millimetre sized oil droplet walking on a vertically vibrated bath of the same fluid provides a macroscopic analogue of pilot-wave dynamics. We report single and double-slit diffraction measurements using an automated trajectory tracking system, collecting over 5000 filtered trajectories across five single-slit widths (10–22 mm) and two double-slit configurations (10 and 14 mm). The far-field angular distributions agree quantitatively with the Fraunhofer sinc^2 envelope using the Faraday wavelength $\lambda_F = 5.19$ mm calculated from the dispersion relation, and near-field measurements reveal discrete Fresnel nodal structure. A no-slit control confirmed the absence of diffraction structure without an aperture. Double-slit data shows interference like angular modulation beyond the single-slit envelope, and node crossing trajectories favour de Broglie's double-solution framework over Bohmian mechanics. To our knowledge, these results constitute the first successful replication of Couder and Fort's 2006 slit diffraction findings.



1 Introduction

“The test of all knowledge is experiment. Experiment is the sole judge of scientific ‘truth.’”
— R. P. Feynman [4]

Quantum mechanics is often encountered as a mathematically exact but physically unintuitive framework [5]. Walking oil droplets are millimetric droplets that bounce on a vertically vibrated bath of the same fluid and are guided by the surface waves they generate, forming a striking classical fluid system in which wave-particle coupling produces behaviours reminiscent of quantum phenomena [6]. This system represents the first macroscopic realisation of a pilot-wave-like dynamics, relevant to foundational questions about wave-particle duality [7]. In this project, we developed a reliable experimental setup for single and double-slit diffraction of walking droplets, with automated trajectory tracking. We used the measured trajectories to test whether this system reproduces diffraction-like behaviour, and to assess whether the observed trajectory structure is more consistent with de Broglie’s double-solution framework or with Bohmian mechanics [3].

Pilot-wave hydrodynamics was discovered in 2005 by Yves Couder and Emmanuel Fort and in 2006 they published a paper reporting the findings of diffraction and interference-like statistics from slit experiments [8]. Since then, the broader field of hydrodynamic quantum analogues has reported classical counterparts to tunnelling [9] and quantised orbits in circular corrals [10]. However, the slit diffraction results in particular have proved controversial. Andersen *et al.* [11] challenged the findings on statistical grounds, while Pucci *et al.* [1] found that the key features arose only in a narrow region of parameter space and identified shortcomings in the original experimental setup. A central motivation of this project was to therefore revisit these slit experiments with improved apparatus and substantially larger trajectory datasets.

Building on the apparatus originally developed by Casson and Machin [2], we improved the experimental setup, designed new slit modules, and implemented an automated trajectory tracking pipeline to collect large single and double-slit datasets. The remainder of this report presents the background theory, experimental methods, and results of these measurements.

2 Background and Theory

2.1 Walking Droplets and Pilot-Wave Hydrodynamics

The central phenomenon underpinning this project is the behaviour of a millimetre-sized oil droplet bouncing on the surface of a vibrating bath of the same fluid. When a bath of silicone oil is vibrated vertically with acceleration $\gamma(t) = \gamma_m \cos(\omega t)$, standing waves known as Faraday waves appear on the surface once the acceleration amplitude γ_m exceeds a critical threshold, the Faraday threshold γ_F [7] [6].

In the regime just below this threshold ($\gamma_m < \gamma_F$), the fluid surface remains flat in the absence of perturbations. A droplet placed on the surface can be kept from coalescing by the intervening air layer, bouncing at the driving frequency or its subharmonics. When the droplet bounces at half the driving frequency (period $2T_F$), it resonates with the Faraday waves it generates. This interaction breaks the symmetry of the bouncing, causing the droplet to be propelled by the slope of its self-generated wave field, a state known as a “walker” [6].

The trajectory of the walker is governed by its interaction with the wave field generated by its previous bounces. A crucial parameter in this system is the “memory” (Me), which quantifies how long the waves persist on the surface before decaying. The memory is largely determined by the proximity to the Faraday threshold:

$$Me \propto \frac{1}{1 - \gamma_m/\gamma_F} \quad (1)$$

In this limit, the superposition of waves from many previous bounces creates a complex, spatially extended pilot-wave field [12]. As γ_m approaches γ_F , the waves become increasingly persistent, meaning the droplet’s trajectory depends on an increasingly long history of previous bounces. It is in this high-memory regime that the system exhibits chaotic dynamics and emergent quantum-like statistics [7].

The wave field generated by each bounce is quasi-monochromatic, with Faraday wavelength λ_F prescribed by the subharmonic condition and the water-wave dispersion relation:

$$\left(\frac{\omega}{2}\right)^2 = \left(gk_F + \frac{\sigma k_F^3}{\rho}\right) \tanh(k_F H) \quad (2)$$

where ω is the angular driving frequency, $k_F = 2\pi/\lambda_F$ is the Faraday wavenumber, g is gravitational accelera-

tion, σ the surface tension, ρ the fluid density, and H the bath depth [7]. In the hydrodynamic analogy, the Faraday wavelength plays a role analogous to the de Broglie wavelength.

2.2 De Broglie's Double-Solution and Pilot-Wave Theory

Pilot-wave theory proposes that quantum particles follow definite trajectories, guided by a real wave field. This stands in contrast to the Copenhagen interpretation, in which the wavefunction is a probability amplitude and particle trajectories are not defined between measurements.

In 1927, De Broglie proposed that particles possess an internal vibration and are guided by a real, physical wave: the "pilot wave" [13]. De Broglie's double-solution theory posited two waves: one is a statistical wave (later identified with the Schrödinger wavefunction) and the other is a real, nonlinear, localised wave centred on the particle. The particle is guided by the phase gradient of this real wave [6].

De Broglie's key physical insight, the 'harmony of phases', is that the particle's internal oscillation remains in phase with the extended guiding wave. This resonance condition is what determines the particle's motion. This is directly realised in the walker system, where the droplet bounces in resonance with the Faraday wave field [6].

In 1952, Bohm reformulated pilot-wave theory using the Schrödinger wavefunction as the guiding field [14]. In this formulation, particle trajectories follow the probability current streamlines, which cannot cross symmetry nodes in single-particle diffraction [5].

Although the walking-droplet system is not a literal model of quantum mechanics, since it is classical, dissipative, and externally driven, it provides a controllable macroscopic platform in which wave-mediated trajectory dynamics and quantum-like statistics can be investigated directly.

2.3 Slit Scattering in Walking-Droplet Systems

Single and double-slit diffraction provide one of the clearest tests of wave-mediated particle behaviour, since individual trajectories are localised but the ensemble distribution may exhibit structured diffraction or interference-like statistics.

When a walker approaches a slit, its pilot wave interacts with the slit boundaries. In the stroboscopic model [15],

the droplet's horizontal motion is governed by:

$$m\ddot{X}_p + D\dot{X}_p = -mg\nabla H(X_p, t) \quad (3)$$

where m is the droplet mass, X_p is its horizontal position, D is the effective drag coefficient arising from the droplet's interaction with the bath surface, and $H(X_p, t)$ is the total wave field height at the droplet location, given by a superposition of waves from all previous bounces [7]. The factor g reflects that the wave slope exerts a force through the effective gravitational acceleration during the bouncing interaction. The slit boundaries modify this wave field, creating an effective transient potential that deflects the droplet's trajectory. Pucci *et al.* [1] showed that the barriers can be modelled as secondary wave sources, and that the resulting walker distributions exhibit near-field peaks whose number increases with slit width, in qualitative agreement with quantum mechanics [16].

Couder and Fort (2006) first reported single and double-slit diffraction-like statistics for walking droplets [8], but subsequent studies highlighted the sensitivity of these results to experimental parameters [11] [1].

Darrow and Bush (2025) demonstrated Fraunhofer diffraction in a theoretical classical pilot-wave model, showing that deterministic, local particle-wave coupling can produce sinc^2 statistics. Notably, their predicted trajectories cross symmetry nodes, unlike Bohmian predictions [3].

Previous studies have shown that walking-droplet slit experiments are highly sensitive to control parameters and boundary effects, making new precise measurements valuable.

2.4 Fraunhofer and Fresnel Diffraction Theory

In the far-field (Fraunhofer) limit, the intensity distribution for a monochromatic plane wave diffracted by a single slit of width a is given by the standard sinc^2 envelope:

$$I(\theta) \propto \text{sinc}^2 \left(\frac{\pi a \sin \theta}{\lambda_F} \right) \quad (4)$$

where θ is the deflection angle.

For the double-slit geometry, with two slits of width a separated by centre-to-centre distance d , the intensity pattern is modulated by a cosine interference term:

$$I(\theta) \propto \cos^2 \left(\frac{\pi d \sin \theta}{\lambda_F} \right) \text{sinc}^2 \left(\frac{\pi a \sin \theta}{\lambda_F} \right) \quad (5)$$

The sinc^2 term is the single-slit diffraction envelope and the $\cos^2$ term encodes the interference between the two apertures.

The Fresnel number is the parameter that distinguishes near-field and far-field regimes:

$$F_N = \frac{a^2}{\lambda_F L} \quad (6)$$

where L is the distance from the slit to the observation point. When $F_N \ll 1$, the pattern is in the Fraunhofer regime (angle-defined, with a sinc^2 envelope); when $F_N \gtrsim 1$, the pattern is in the Fresnel regime (position-dependent, with nodal structure set by the Fresnel–Kirchhoff integral [17]). Our measurements span both regimes.

In the Fresnel regime, the transverse position distribution immediately after the slit exhibits nodal structure, whereas in the Fraunhofer regime the pattern is characterised by the angular distribution of trajectories in the far field.

In the walker system, these expressions are applied with the Faraday wavelength λ_F replacing the optical or de Broglie wavelength, and the statistical distribution of deflection angles over many trajectories replacing the wave intensity.

3 Experimental Method

3.1 Apparatus Overview

This project builds upon a walking-droplet apparatus originally developed in the previous academic year by Grace Casson and Ben Machin [2]. Key modifications made during this project include new 3D-printed slit modules, a re-designed launcher adapted from Pucci et al. (2017) [1], automated droplet generation, and full recalibration.

The experiment employs a shallow bath of 20 cSt silicone oil mounted on the cone of a vertically driven Fane Sovereign 12300 loudspeaker (Figure 1). The speaker is driven sinusoidally at a frequency of 70.5 Hz with a voltage of 14.0 Vpp. The bath was operated in the high-memory regime, with $\gamma/\gamma_F = 0.936 \pm 0.015$, where γ_F was determined by increasing the driving amplitude until the onset of surface instability. The bath was housed in a plastic enclosure to shield from air currents, with LED illumination around the bath edge to improve droplet contrast in the overhead imaging system [1]. The droplets are generated by a relay-driven wire arm which is controlled by an Arduino. Periodic interruption of the drive current

causes the wire to briefly contact the oil surface, producing millimetre-sized droplets in a reproducible manner. The geometry of the bath, slit barriers, and oil depth are shown in Figures 2, 3, and 4.

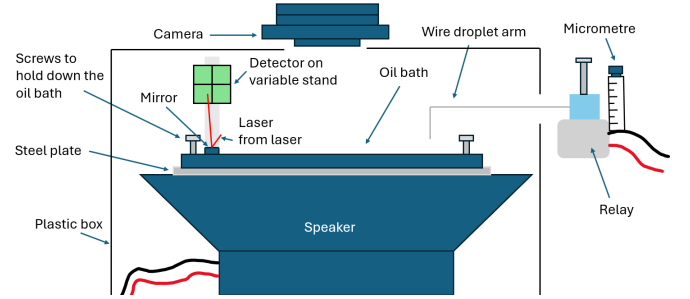


Figure 1: Side-view schematic of the experimental apparatus. A circular silicone-oil bath is mounted on a Fane Sovereign 12300 loudspeaker driven at 70.5 Hz with screws. Droplets are generated by a wire arm whose vertical displacement is controlled by a relay and micrometre stage. Bath displacement is monitored by reflecting a laser beam off a mirror mounted on the bath surface onto a quadrant photodetector, used to calibrate the forcing acceleration γ . A steel plate beneath the bath provides a magnetic mounting surface for the 3D-printed slit modules. The entire assembly is enclosed in a plastic box to suppress air currents. Trajectories are recorded by an overhead camera (see Figure 2 for the camera view).

3.2 Achieving Stable Walkers

Stable walking required precise control of several interdependent parameters. The bath acceleration had to be maintained just below the Faraday threshold γ_F , since the memory of the system diverges sharply as this limit approaches (Equation 1). Small changes in forcing amplitude could shift the droplet from stable walking into either a purely bouncing or chaotic regime.

The Faraday threshold itself was sensitive to temperature, since the viscosity and surface tension of silicone oil are temperature-dependent. The bath temperature was monitored throughout each session and typically ranged from 26–30 °C; a warm-up period was required before stable walking could be achieved. The system was also sensitive to the oil depth: changes as small as 0.1 mm affected the slit-well formation and the resulting diffraction patterns.

To minimise environmental disturbance, the plastic enclosure was kept sealed during data collection, and movement in the room was minimised. To ensure that each trajectory was independent of previous runs, the driving signal was interrupted between droplet launches, allowing the residual wave field to decay fully before the next

droplet was generated. The wave generated by the relay strike was measured by video to propagate at approximately 1.5 m s^{-1} (15 mm in 10 ms), far exceeding the walker speed ($8\text{--}14 \text{ mm s}^{-1}$). This confirmed that the disturbance had exited the slit region before the droplet began walking. The relay then produced a new droplet onto a quiescent bath surface. This cycling procedure was later automated using a two-Arduino system (Section 3.3). This also prevented the accumulation of stationary droplets at the bath boundary, whose residual wave fields could otherwise bias subsequent trajectories.

3.3 Slit Module and Launcher Design

The previous launcher design [2] suffered from oil leakage leading to inconsistent fluid depths, and launched droplets at excessive speed, giving insufficient time for the walker to stabilise before reaching the slit. A new solid 3D-printed launcher was designed in Autodesk Inventor, adapted from the design of Pucci et al. [1]. This launcher guides the droplet from the point of creation to the slit, allowing it to reach a stable walking speed before encountering the barrier geometry. The droplet generation cycle was automated using an Arduino-controlled relay, with modified timing (relay on for 7 s, off for 2 s) to reduce ripple interference from the relay strike. This was later upgraded to a two-Arduino system controlling both the speaker drive and droplet generation, enabling fully automated cycling (Appendix G).

The inherited apparatus also used barriers that were not fully submerged, which could introduce spurious boundary effects not representative of the wave-mediated scattering studied in the literature [1]. Following standard practice in the field, a new modular submerged barrier system was developed. The slit assemblies consisted of 3D-printed carriers with embedded magnets for secure placement on the steel bath plate, and handles allowing the barriers to be positioned reproducibly while remaining fully submerged. The slit assemblies consisted of 3D-printed carriers with embedded magnets for secure placement on the steel bath plate, and handles allowing the barriers to be positioned reproducibly while remaining fully submerged. A removable central insert allowed quick switching between single and double-slit configurations. The double-slit centre piece was initially cuboid, but was replaced with 3 mm diameter circular magnets to avoid trapping droplets behind the barrier. The barrier height was uniform across all slit modules (4 mm); the step in fluid depth above the barriers increases the local Faraday threshold, forming an effective hydrodynamic potential barrier (Figure 3).

Single-slit widths (L) of 10, 12, 17, 19 and 22 mm were

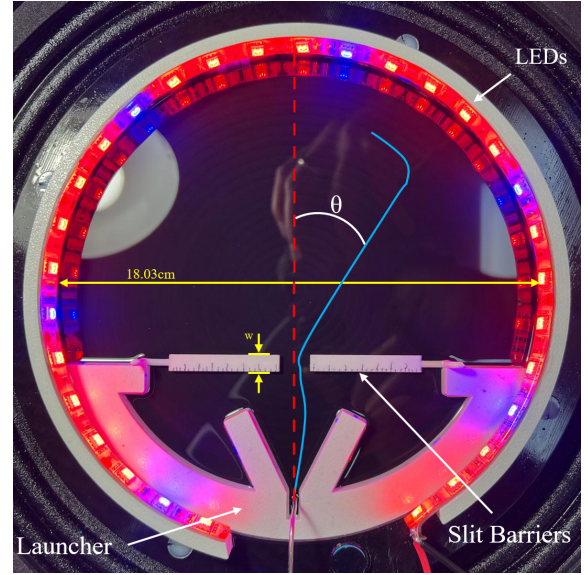


Figure 2: Bird's-eye camera view of the experimental apparatus in the single-slit configuration. The oil bath (diameter 18.03 cm) is illuminated by a ring of LEDs to enhance droplet visibility for overhead tracking. The 3D-printed slit barriers ($w = 0.625 \text{ cm}$) are visible either side of the slit opening. A representative tracked trajectory (light blue) shows the characteristic deflection through the slit region, with the scattering angle θ measured from the slit centre line (red dashed) as used throughout the Report.

tested. For double-slit diffraction, slit widths (L) of 14 mm ($d=17 \text{ mm}$) and 10 mm ($d=13 \text{ mm}$) were used, where d is the centre-to-centre slit separation. Full CAD drawings and design iterations are included in Appendix H.

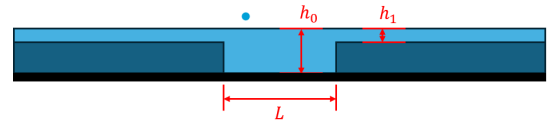


Figure 3: Cross-section of the single-slit barrier geometry. The blue circle represents the walking droplet. The fluid depth above the barrier is h_1 and the fluid depth to the bath floor is h_0 , with L denoting the slit width. The reduction in depth over the barrier increases the local Faraday threshold, creating an effective hydrodynamic potential that deflects walker trajectories. Adapted from [1].

3.4 Calibration

The laser displacement sensor was calibrated to convert the detector's voltage output into an absolute vertical displacement of the vibrating bath. The displacement detector (Hamamatsu S5990-01 PSD) consisted of a quadrant photodiode onto which a laser beam, reflected from

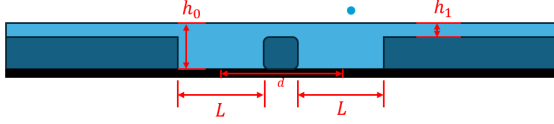


Figure 4: Cross-section of the double-slit geometry. The blue circle represents the walking droplet. The two slit openings of width L are separated by a central barrier, with d denoting the separation between slit centrelines and fluid depths h_0 and h_1 indicated. Adapted from [1].

a marker (mirror) attached to the bath, was directed. The differential output along one axis ($y_1 - y_2$) was monitored on an oscilloscope.

To calibrate it, the detector was moved vertically in small increments using a micrometre stage and the mean (DC) voltage was recorded at each height. Repeating this from the bottom of the active area until the beam moved off the detector gave a near-linear relationship between voltage and height (Figure 5). The calibration was linear over the working range, with gradient $k = -19.49 \pm 0.22 \text{ mV mm}^{-1}$ ($R^2 \approx 0.999$).

Once this factor was known, the speaker was driven and the peak-to-peak voltage V_{pp} was measured on the oscilloscope, then converted into a peak-to-peak displacement of the bath:

$$y_{pp} = \frac{V_{pp}}{dV/dy} \quad (7)$$

The bath acceleration was then obtained from

$$\gamma = y_{pp} (2\pi f)^2 \quad (8)$$

where $f = 70.5 \text{ Hz}$ is the driving frequency. Uncertainties were calculated from the fit error on the calibration gradient and the oscilloscope voltage resolution (Appendix B). This procedure provides an absolute measure of bath displacement independent of amplifier settings, enabling consistent identification of operating conditions relative to the Faraday threshold across sessions.

The displacement detector was recalibrated after the final apparatus configuration was established, including the redesigned launcher and fixed operating parameters; the resulting gradient ($k = -19.49 \pm 0.22 \text{ mV/mm}$) replaced the preliminary calibration reported in the interim report.

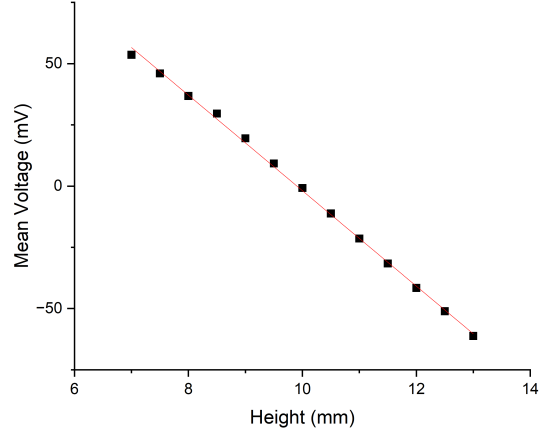


Figure 5: Calibration of the laser displacement detector. Mean (DC) photodiode output voltage V (mV) plotted against detector height y (mm), measured by translating the detector vertically using a micrometre stage. The linear fit gives a calibration gradient $k = dV/dy = -19.49 \pm 0.22 \text{ mV/mm}$ ($R^2 = 0.999$), used to convert the observed peak-to-peak voltage V_{pp} into bath displacement.

3.5 Image Acquisition and Trajectory Analysis

A Prosilica GC1290 monochrome camera was mounted above the bath via a clamp stand, recording through a hole in the plastic enclosure to provide a stable top-down view of the slit region. Directed LED lighting around the bath perimeter was required to achieve reliable droplet contrast against the background wave field. The spatial calibration was 0.114 mm/px , giving a field of view of approximately $73 \times 38 \text{ mm}$. The frame rate reported by the Vimba acquisition software (32.8 fps) was found to yield unrealistic droplet speeds of $\sim 40 \text{ mm/s}$; an independent timing measurement, in which the elapsed time for a known number of acquired frames was recorded within the Python acquisition script, gave 9.17 fps, producing physically consistent walker speeds. This corrected value was used throughout the analysis.

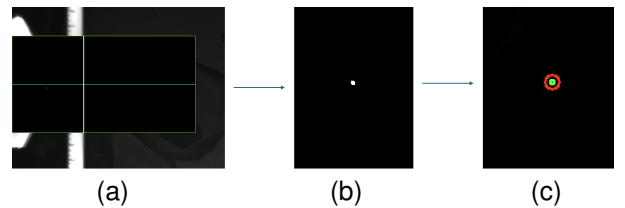


Figure 6: Stages of the automated tracking pipeline. (a) Raw camera frame with region of interest indicated. (b) Binary thresholded image isolating the droplet. (c) Centroid detection used to extract the droplet position per frame.

An automated tracking pipeline was developed in Python using OpenCV and the Allied Vision `vimbo` interface. Within this field of view, droplets were identified using adaptive thresholding and geometric filtering (circularity, aspect ratio, and area constraints), and their centroids were computed from detected contours (Figure 6). Temporal continuity constraints were imposed such that detections were preferentially accepted near the previous droplet position, reducing false identifications and short drop-outs. Each completed trajectory was saved as a time-ordered sequence of positions in a long-format CSV file, with one file per acquisition run. This format allowed trajectory sets from multiple acquisition runs to be combined in post-processing, providing robustness against interruptions such as camera displacement. The full tracking algorithm and parameter values are documented in Appendix A, with all code available via the linked repository.

From each retained trajectory, two scattering parameters were extracted. The impact parameter y_i was defined as the transverse position at which the droplet crossed the slit exit line, obtained by linear interpolation between adjacent trajectory points. The far-field deflection angle θ was determined by identifying a plateau region in the local angle measured over sliding windows downstream of the slit; where no stable plateau was found, a fixed two-point measurement was used as a fallback (see Appendix A).

Trajectories were filtered to retain only those with: mean pre-slit speed within a slit-width-dependent window (typically 8–14 mm/s; see Appendix D and Figure 7 for visualisation), a minimum track length of 25 points, and a maximum absolute deflection angle of 75° . Droplets below 8 mm/s were typically too slow to pass through the slit cleanly and tended to become trapped at the boundaries. Trajectories with deflection angles exceeding 75° were also excluded, as these were typically associated with edge interactions rather than genuine diffraction. This filtering excluded approximately 30 – 50% of raw trajectories, depending on slit width and experimental conditions. The implications of this filtering for potential selection bias are addressed in Section 6.

For double-slit experiments, the same tracking and filtering pipeline was applied and each trajectory was classified by which slit it had passed through, based on its transverse position relative to the central divider at the point of slit entry.

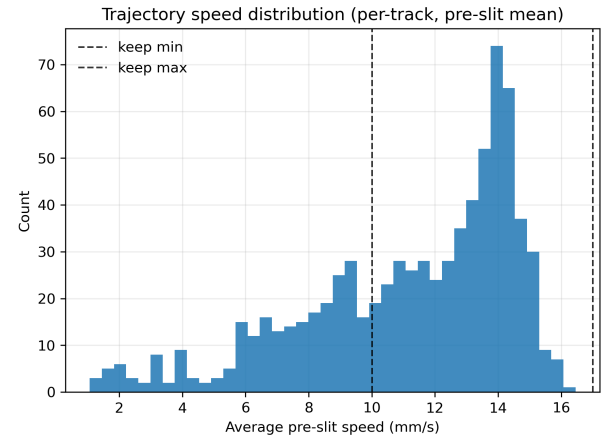


Figure 7: Distribution of mean pre-slit speeds for all tracked trajectories (17 mm slit width, $N = 912$). The dominant peak corresponds to walkers at the expected free-walking speed. Slower droplets, typically produced by variability in the launcher, tend to interact with the slit boundaries rather than passing through cleanly. Dashed lines indicate the speed filtering window used to retain well-behaved walkers for the angular distribution analysis.

4 Results

4.1 Single-Slit Trajectories and Diffraction-Like Broadening

Single-slit experiments were conducted for slit widths of 10, 12, 17, 19 and 22 mm. In all cases, the post-slit trajectories exhibited a symmetric diffractive fan about the centreline, with narrower slits producing broader angular distributions as expected from single-slit diffraction theory (see also Appendix J for the theoretical envelopes across all slit widths). The 19 mm dataset is presented in Appendix C and discussed in Section 5.

Figure 9 shows the 12 mm case, which yielded the cleanest dataset with $N = 395$ retained trajectories from 633 raw trajectories. Droplets enter from the left, pass through the slit, and spread into a characteristic fan. The transverse density profile at $x = 40$ mm downstream reveals a central peak with weaker intensity at larger deflection angles, consistent with wave-mediated diffraction. For comparison, the optical single-slit pattern produced by a 635 nm diode laser (Figure 9, right) exhibits the same qualitative structure.

The deflection angle distributions for four slit widths are shown in Figure 8, overlaid with the Fraunhofer single-slit intensity envelope (Equation 4), where a is the slit width and $\lambda_F = 5.19$ mm is the Faraday wavelength calculated from the dispersion relation (Equation 2). Each theoretical curve was centred on the median of the measured

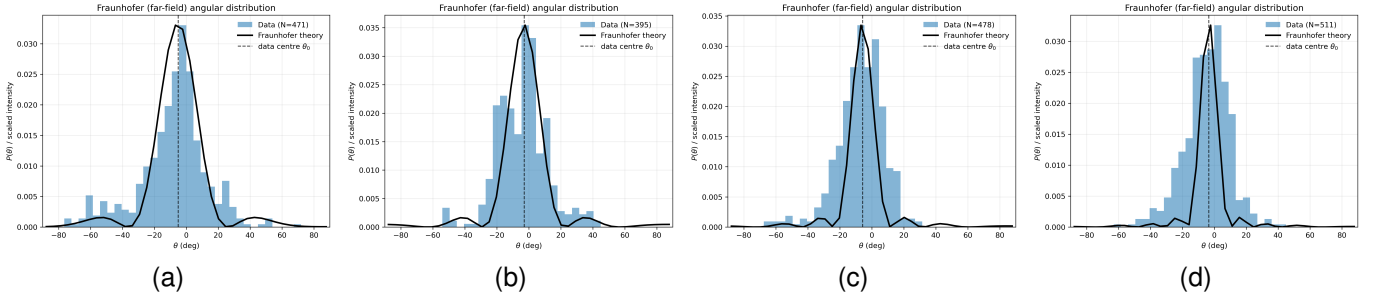


Figure 8: Deflection angle distributions for single-slit widths of (a) 10 mm ($N = 471$), (b) 12 mm ($N = 395$), (c) 17 mm ($N = 478$) and (d) 22 mm ($N = 511$), overlaid with the Fraunhofer sinc^2 envelope (Equation 4) using $\lambda_F = 5.19$ mm. For the 10 mm slit, a larger combined dataset ($N = 471$) was used to improve statistical resolution. Narrower slits produce broader distributions with first minima at larger θ . Subsidiary maxima are visible flanking the central peak in all four cases.

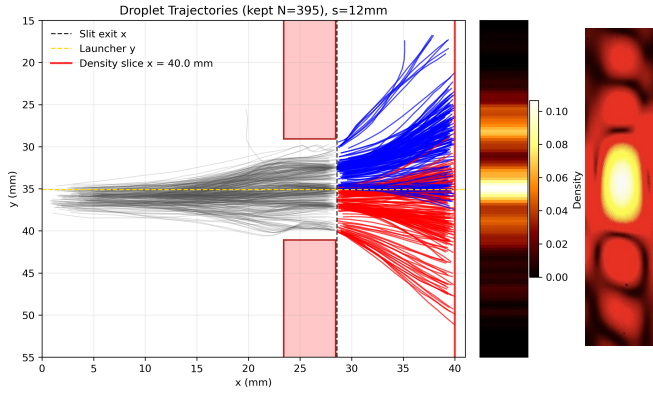


Figure 9: Left: retained trajectories ($N = 395$) through the 12 mm single slit, coloured by entry side (blue/red) with pre-slit paths in grey. The colour strip at $x = 40$ mm shows the transverse intensity profile. Right: optical single-slit diffraction pattern from a 635 nm diode laser, showing the analogous central maximum and subsidiary fringes.

distribution to correct for a small systematic angular offset; no other free parameters were used.

As a decreases from 22 to 10 mm, the central maximum broadens and the first minima move to larger θ , precisely as predicted by Equation 4. Subsidiary maxima flanking the central peak are resolved for all four slit widths, providing further confirmation of the sinc^2 form. The 12 mm case shows the closest agreement between theory and data. For wider slits (17 and 22 mm), the observed distributions are broader than the theoretical envelope; this is consistent with broadening from the spread of droplet speeds and launch conditions, and is discussed further in Section 5.

4.2 Fresnel and Fraunhofer Regimes

The Fresnel number (Equation 6) determines whether the transverse distribution is position-dependent ($F_N \gtrsim 1$, Fresnel regime) or angle-defined ($F_N \ll 1$, Fraunhofer

regime). At the near field measurement position ($z = 2.29$ mm after the slit), the Fresnel numbers range from $F_N \approx 8$ for the 10 mm slit to $F_N \approx 41$ for the 22 mm slit, placing all measurements well within the Fresnel regime. At the far-field measurement distance ($z \approx 11.4$ mm), these values drop to $F_N \approx 1.7$ – 8.2 , approaching the Fraunhofer regime where the angular sinc^2 pattern of Section 4.1 is recovered.

Figure 10 (b,f) shows the transverse position distribution at $z = 2.29$ mm for the 12 mm slit, overlaid with the Fresnel–Kirchhoff diffraction integral evaluated using $\lambda_F = 5.19$ mm. The near-field distribution exhibits discrete preferred positions: three central peaks and two edge features associated with the slit boundaries are resolved in both the data and the theoretical prediction. This spatial structure is also visible in the trajectory plot (Figure 10), where the post-slit paths cluster into distinct bands near the slit exit before diverging into the far-field fan.

Figure 10 shows the near-field patterns for four slit widths alongside the corresponding trajectory plots zoomed to the slit-exit region. As the slit width increases from 12 to 22 mm, the number of predicted Fresnel oscillations increases in proportion to the Fresnel number. The 12 mm case resolves the nodal structure most clearly; at 17 mm, additional oscillations appear and remain partially resolved; at 22 mm, the oscillations become too closely spaced to resolve with the available statistics.

4.3 Faraday Wavelength

The Faraday wavelength used for all theoretical overlays was calculated from the dispersion relation (Equation 2) using the experimental parameters: driving frequency $f = 70.5$ Hz, oil density $\rho = 950$ kg m $^{-3}$, surface tension $\sigma = 19.94$ mN m $^{-1}$ (at 28 °C), and bath depth

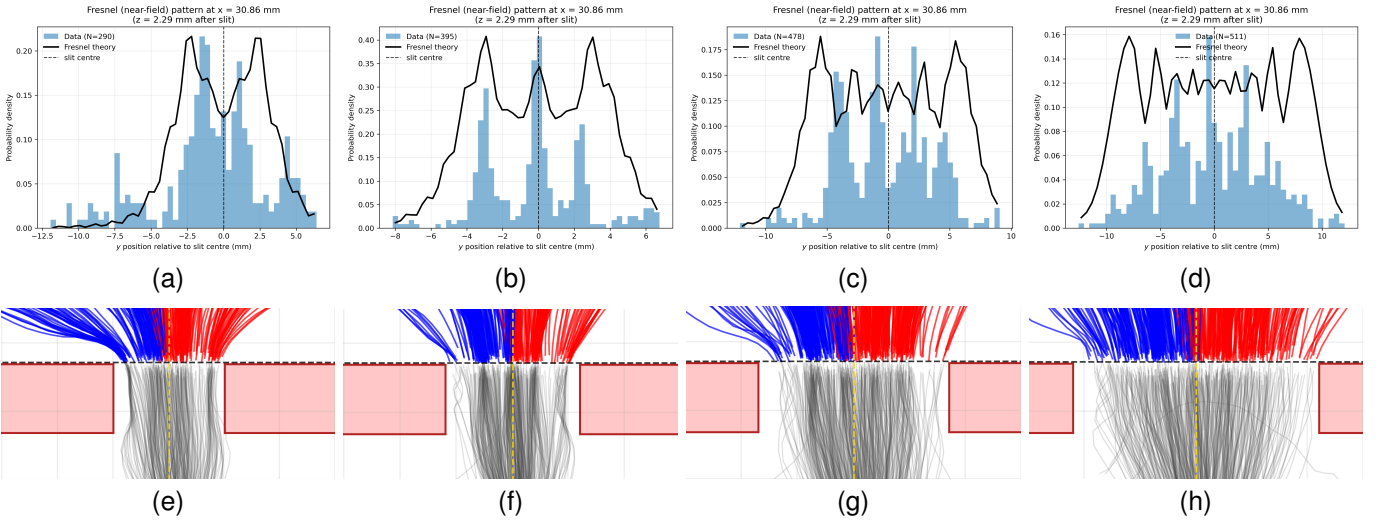


Figure 10: Fresnel diffraction patterns and near-field trajectory groupings for single-slit widths of 10 mm (a, e), 12 mm (b, f), 17 mm (c, g) and 22 mm (d, h). Top row: transverse position distributions at $z = 2.29$ mm after the slit, overlaid with the Fresnel–Kirchhoff integral prediction (solid line). As slit width increases, the number of predicted Fresnel fringes increases and the structure becomes progressively harder to resolve at the available statistics. Bottom row: zoomed trajectories near the slit exit, coloured blue/red by entry side. Discrete preferred paths are visible, particularly for narrower slits. The trajectory counts for each slit width are: 10 mm ($N = 290$), 12 mm ($N = 395$), 17 mm ($N = 478$), 22 mm ($N = 511$).

$H = 5$ mm. The oil properties are manufacturer-specified values for 20 cSt silicone oil [18], with the surface tension corrected from 20.3 mN m^{-1} at 22°C using the linear temperature coefficient $\partial\sigma/\partial T = -0.06 \text{ mN m}^{-1} \text{ K}^{-1}$. This gives $\lambda_F = 5.19 \pm 0.05$ mm. This single value was used consistently across all slit widths for both the Fraunhofer and Fresnel theoretical predictions; no per-dataset fitting of the wavelength was performed. The full calculation is documented in Appendix I.

4.4 Node-Crossing Trajectories

Figure 11 shows the deflection angle θ plotted against the impact parameter y_i for the 12 mm slit. A clear correlation is visible: droplets entering above the slit centre ($y_i > 0$) are preferentially deflected upward, and those entering below ($y_i < 0$) are deflected downward. However, the spread about this trend is substantial, and a number of trajectories are observed to cross one another, with droplets starting at similar impact parameters exiting at widely different angles. This node-crossing behaviour is also visible directly in the trajectory plots of Figures 9 and 13, where blue and red paths (coloured by entry side) can be seen crossing the slit centre line after the slit exit.

For comparison, Figure 12 shows the theoretical diffraction maps calculated by Darrow and Bush [3] for (a) Bohmian and (b) de Broglie pilot-wave models. In Bohmian mechanics, trajectories follow probability cur-

rent streamlines and cannot cross symmetry nodes [5]; the resulting diffraction map is a smooth, monotonic function. In the de Broglie model, trajectories exhibit a chaotic fold structure with frequent node crossings, closely resembling the experimental scatter observed in Figure 11. The implications of this comparison are discussed in Section 5.

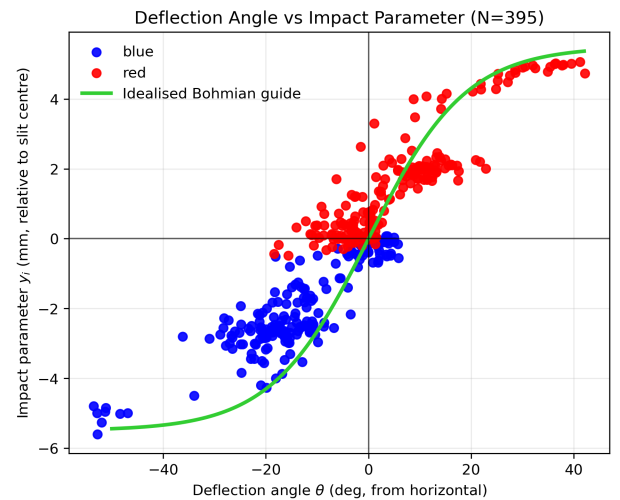


Figure 11: Deflection angle θ versus impact parameter y_i for the 12 mm single slit ($N = 395$), coloured by entry side (blue: above slit centre, red: below). The overall correlation between entry position and exit angle is consistent with pilot-wave-mediated deflection. The scatter about any monotonic trend indicates trajectory crossing, inconsistent with Bohmian mechanics (see Figure 12). The green curve represents the Bohmian prediction, in which trajectories cannot cross the centreline [3].

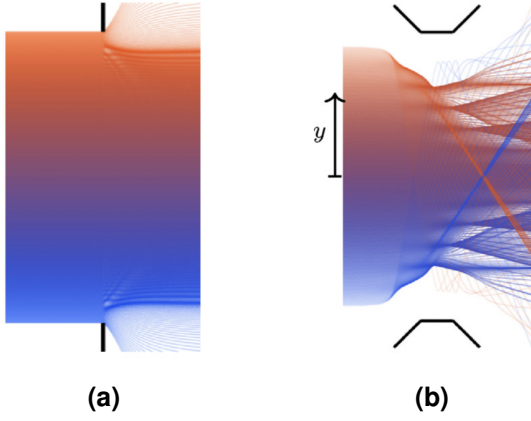


Figure 12: Theoretical single-slit diffraction maps from Darrow and Bush [3], showing impact parameter y_{in} versus diffraction angle θ_{out} . (a) Bohmian trajectories: the map is smooth and monotonic, with no node crossing. (b) De Broglie pilot-wave model (coupling parameter $b = 16.7$): the map exhibits a chaotic fold structure with frequent node crossings, qualitatively consistent with the experimental data in Figure 11. Taken from Ref. [3], Figs. 5(a) and 5(c).

4.5 Double-Slit Results

Double-slit experiments were conducted with slit width $L = 10$ mm and centre-to-centre separation $d = 13$ mm, using a 3 mm diameter circular centre piece to divide the aperture. Figure 13 shows the $N = 1252$ retained trajectories, coloured by which slit the droplet passed through based on its transverse position relative to the centre piece.

The deflection angle distribution (Figure 13) is overlaid with the double-slit intensity formula (Equation 5) using $L = 10$ mm, $d = 13$ mm, and $\lambda_F = 5.19$ mm, with the peak position shifted to account for the systematic angular offset as in the single-slit case. The distribution shows a clear central maximum with subsidiary peaks on either side, consistent with the interference modulation predicted by the $\cos^2$ term in Equation 5. The raw distribution exhibited a systematic bias toward positive deflection angles, consistent with the apparatus asymmetry also observed in the single-slit data.

The double-slit results are more sensitive to experimental conditions than the single-slit case, and the statistical scatter is greater. Nevertheless, the presence of interference-like modulation in the angular distribution, which is absent in the single-slit data for the same individual slit width, indicates that the second aperture influences the trajectory statistics through the shared pilot-wave field.

A second double-slit configuration was also investigated, with slit width $L = 14$ mm and centre-to-centre

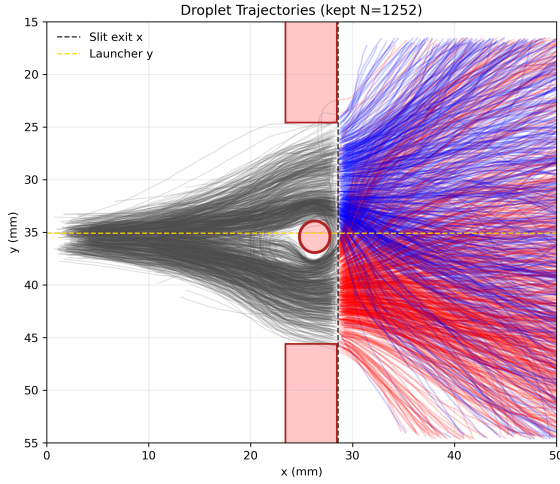
separation $d = 17$ mm, yielding $N = 830$ retained trajectories (Appendix F). The angular distribution for this wider geometry showed qualitatively similar interference-like modulation, with the peak spacing narrowing as expected for larger d .

5 Discussion

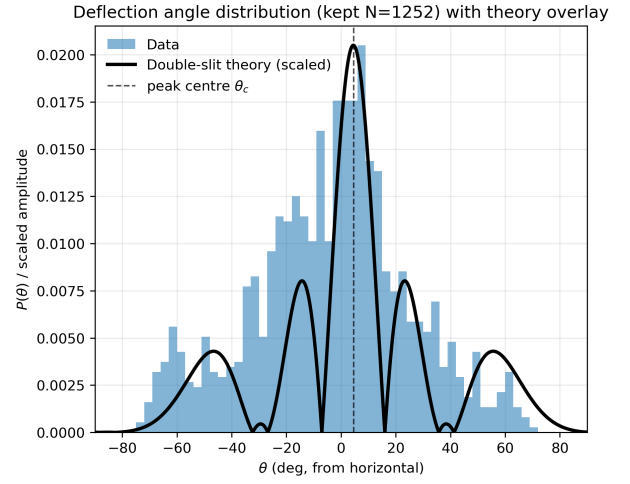
5.1 Summary of Key Findings

The single-slit results demonstrate reproducible diffraction-like angular broadening across five slit widths and multiple experimental sessions. The Fraunhofer sinc^2 envelope (Equation 4) captures the width of the central maximum, the angular positions of the first minima, and the subsidiary maxima for all four slit widths presented, using a single, independently calculated Faraday wavelength $\lambda_F = 5.19$ mm with no per-dataset fitting. The predicted trend, that narrower slits produce broader angular distributions with first minima at larger $|\theta|$, is confirmed quantitatively across the 10–22 mm range. The 12 mm case shows the closest agreement between theory and data. For wider slits (17 and 22 mm), the observed distributions are broader than the theoretical prediction. The sinc^2 envelope assumes that all droplets scatter with the same effective wavelength, but in practice the droplets span a range of speeds, and slower walkers produce broader angular distributions than faster ones. Averaging over this speed spread smears the diffraction pattern, broadening it relative to the monochromatic prediction, an effect analogous to chromatic aberration in optics, where a spread of wavelengths blurs an otherwise sharp image. Darrow and Bush [3] account for a similar effect in their theoretical model by convolving the Fraunhofer prediction with a Gaussian of width proportional to the initial-state uncertainty (their Equation 4), which broadens the predicted distribution in a manner consistent with the speed-dependent smearing observed here. The correlation between droplet speed and impact parameter (Appendix D) supports this interpretation.

The near-field measurements reveal the expected Fresnel-to-Fraunhofer crossover. At $z = 2.29$ mm after the slit, the transverse position distributions show discrete preferred positions whose spacing and number are consistent with the Fresnel–Kirchhoff diffraction integral evaluated using the same $\lambda_F = 5.19$ mm. For the 12 mm slit, three central peaks and two edge features associated with the slit boundaries are resolved in both the data and the theoretical prediction (Figure 10). As the slit width



(a)



(b)

Figure 13: (a) Double-slit trajectories ($N = 1252$) with slit width $L = 10$ mm and centre-to-centre separation $d = 13$ mm, coloured by slit passage (blue/red). The circular centre piece is indicated. (b) Deflection angle distribution overlaid with the double-slit intensity formula (Equation 5) using $\lambda_F = 5.19$ mm. Interference-like peak structure is visible on both sides of the central maximum.

increases, the number of predicted Fresnel oscillations grows in proportion to the Fresnel number, and this progression is observed experimentally, though the finer oscillations at wider slit widths become increasingly difficult to resolve with the available statistics (Figure 10). Further downstream, this positional structure gives way to the angle-defined sinc^2 pattern, completing the Fresnel-to-Fraunhofer transition.

The double-slit angular distribution exhibits interference-like peaks that are absent in the single-slit data for the same individual slit width of 10 mm. If the second aperture had no influence on the droplet, the double-slit distribution would simply reproduce the single-slit sinc^2 envelope. Instead, subsidiary peaks appear at angular positions consistent with the $\cos^2$ interference term in Equation 5, indicating that the pilot wave passing through the second slit deflects the droplet even though it only passes through one. This is the defining signature of wave-mediated interference in this system.

5.2 Comparison with Literature

The single-slit results presented here reproduce and extend the original findings of Couder and Fort [8], who first reported diffraction-like statistics for walking droplets passing through single and double slits. Their study used approximately 75 trajectories per configuration. Couder and Fort also reflected each trajectory about the slit centreline to double their effective sample size, exploiting the geometric symmetry of the slit. This work retains be-

tween 290 and 511 trajectories per slit width across five systematically varied apertures, with Fraunhofer theory overlays confirming quantitative agreement. Additional datasets containing over 1000 trajectories per slit width were also collected and show consistent diffraction-like patterns, though with greater launch asymmetry. To our knowledge, this represents one of the first successful replications of their slit diffraction results since the original 2006 publication.

Andersen et al. [11] were unable to reproduce coherent interference fringes in a double-slit geometry and questioned whether the original observations were robust. The broader reproducibility debate [19] raised concerns about the sensitivity of walking-droplet experiments more generally. Our double-slit data does show interference-like modulation consistent with the $\cos^2$ term in Equation 5, though with greater statistical noise than the single-slit case. The single-slit results also required careful control of forcing amplitude, oil depth, and temperature to obtain clean diffraction patterns. The sensitivity to these conditions is consistent with the concerns raised by Andersen et al., and may explain why different groups have obtained different outcomes.

Pucci et al. [1] demonstrated that walking-droplet slit experiments are highly sensitive to control parameters and boundary effects, and showed that the slit barriers act as secondary wave sources producing near-field structure whose complexity increases with slit width. Our Fresnel measurements are consistent with this picture. The oil depth over the submerged barriers proved to be a critical parameter: when the oil was too shallow, the Faraday

waves generated at the slit edges extended further into the aperture, strengthening the boundary confinement, effectively narrowing the slit, and causing slower droplets to become trapped at the edges. When the oil was too deep, this boundary confinement weakened and the near-field Fresnel structure was washed out. Changes of order 0.1 mm in oil depth were sufficient to visibly alter the near-field pattern, confirming the sensitivity reported by Pucci et al. [1] and highlighting the precision required for reproducible measurements.

Darrow and Bush [3] demonstrated Fraunhofer diffraction in a theoretical classical pilot-wave model, predicting both sinc^2 statistics and node-crossing trajectories. Our experimental results are in qualitative agreement on both counts: the far-field angular distributions follow the sinc^2 envelope, and the observed trajectories cross symmetry nodes as their model predicts. This agreement between a first-principles theoretical model and independent experimental data strengthens the case that deterministic pilot-wave coupling can produce diffraction-like statistics.

5.3 De Broglie versus Bohm

The walking-droplet system involves a real, physical wave field that guides the particle through the slit geometry. This makes it conceptually closer to de Broglie's double-solution pilot-wave theory [13], in which a particle is guided by a real, nonlinear wave centred on it, than to Bohmian mechanics [14], in which trajectories are determined by the probability current derived from the Schrödinger wavefunction [6].

A key distinction between these two frameworks concerns symmetry nodes. In Bohmian mechanics, single-particle trajectories follow the probability current streamlines, which cannot cross the symmetry axis in a single-slit diffraction geometry [5]. A droplet entering above the slit centre must exit above it; trajectories from opposite sides of the slit cannot cross. In de Broglie's double-solution picture, no such constraint applies: the guiding wave is a real physical entity and the particle dynamics are not bound by the non-crossing theorem that governs Bohmian streamlines [3].

The experimental data presented in Section 4.4 show that trajectories cross one another, with droplets starting at similar impact parameters exiting at widely different angles. The θ versus y_i scatter plot (Figure 11) shows substantial spread about any monotonic trend, inconsistent with the smooth, non-crossing diffraction map predicted

by Bohmian mechanics. This is consistent with the theoretical trajectories of Darrow and Bush [3], whose pilot-wave model produces a chaotic fold structure with frequent trajectory crossings (Figure 12b), and inconsistent with the Bohmian prediction of strictly non-crossing paths (Figure 12a).

This observation should be interpreted with care. The walking-droplet system is a classical, macroscopic, dissipative analogue, not a quantum system. The node-crossing result does not prove that quantum particles follow de Broglie-style trajectories. What it does demonstrate is that a physically realised pilot-wave system, in which a real wave guides a real particle, naturally produces trajectory statistics and crossing behaviour consistent with de Broglie's framework and inconsistent with the Bohmian prediction. The analogy thus provides a concrete, experimentally accessible illustration of how de Broglie-style dynamics can give rise to diffraction-like statistics without requiring the non-crossing constraint of Bohmian mechanics.

5.4 Limits of the Analogue

The walking-droplet system is a macroscopic, classical, dissipative system driven by an external vibration. It is not a quantum system, and the analogy with quantum mechanics has clear boundaries. The Faraday waves decay over time, unlike quantum wavefunctions, and the system involves a single macroscopic particle interacting with its own wave field. Multi-particle quantum phenomena such as entanglement cannot be reproduced. However, other quantum analogues beyond slit diffraction have been demonstrated in similar systems, including tunnelling across submerged barriers [9] and quantised orbital statistics in circular corrals [10], suggesting that the scope of the analogy extends further than the experiments reported here.

Within these boundaries, the results of this work operate at different levels of quantitative strength. The single-slit Fraunhofer distributions and the near-field Fresnel nodal structure both show quantitative agreement with classical diffraction theory evaluated using independently measured parameters. The double-slit results show qualitative agreement, with interference-like modulation present but less statistically definitive. The node-crossing trajectories provide qualitative evidence favouring de Broglie's double-solution framework over Bohmian mechanics, consistent with the theoretical predictions of Darrow and Bush [3], though a rigorous statistical com-

parison remains for future work.

The value of the hydrodynamic pilot-wave system is not as a substitute for quantum mechanics but as a physically accessible platform in which the interplay between particle trajectories and a guiding wave field can be observed directly. In quantum systems, individual trajectories cannot be measured without disturbing the wavefunction; the walker system allows the full trajectory ensemble to be recorded and compared against competing theoretical frameworks such as de Broglie's double-solution and Bohmian mechanics. Equally, understanding where this classical analogue breaks down is itself scientifically informative: the points at which the hydrodynamic system fails to reproduce quantum statistics reveal what is genuinely quantum about quantum diffraction, and what can in principle be captured by deterministic pilot-wave dynamics alone.

6 Future Improvements and Limitations

6.1 Experimental Limitations

The trajectory datasets, while substantial (hundreds per slit width, with additional datasets exceeding 1000), remain small compared to the millions of photons typically used in optical diffraction experiments. The filtering process removes 30–50% of raw trajectories, which could introduce selection bias if the excluded population has systematically different scattering properties. The droplet generator does not produce uniform droplet sizes, resulting in a broad speed distribution that must be filtered to a narrower band, further reducing the usable sample.

The system is sensitive to several parameters that were difficult to control precisely. Oil depth, which was found to affect the near-field Fresnel structure at the 0.1 mm level, also influenced the far-field Fraunhofer distributions: repeat measurements of the 12 mm slit under different oil depths (Figure 17 and 18) showed visibly different angular distributions despite identical slit geometry and forcing conditions. This oil depth could not be measured to the required precision with the available apparatus. Temperature variations between 26 and 30 °C across sessions altered the oil viscosity and surface tension, shifting the Faraday threshold; data collection was restricted to sessions where the bath had reached at least 26 °C, as this was found to produce the most consistent walking behaviour. A residual systematic angular bias was present in some datasets, most likely due to the launcher position

relative to the slit centre rather than a fundamental apparatus asymmetry, as adjusting the launcher position was observed to reduce or eliminate the offset.

6.2 Proposed Improvements

Several practical improvements would strengthen future measurements. A uniform droplet generator producing consistent droplet sizes would narrow the speed distribution, reducing the speed-dependent broadening observed for wider slits and removing the need for aggressive speed filtering [7]. Precise oil-depth measurement, for example using a micrometer gauge or optical method, would allow the sensitivity of the Fresnel structure to oil depth to be characterised quantitatively rather than qualitatively. A temperature-controlled bath environment would stabilise the Faraday threshold between sessions, removing the need to wait for the system to warm up before data collection.

6.3 Scientific Extensions

The apparatus is well suited to several further investigations. Larger double-slit datasets would improve the statistical certainty of the interference-like modulation and allow a more rigorous comparison with theory. Tunnelling experiments using submerged depth steps of varying geometry would test whether the droplet transmission probability depends on barrier width and height in a manner analogous to quantum tunnelling [9]. Circular corral experiments could investigate whether the long-time spatial distribution of a confined walker reproduces the quantised orbital statistics reported by Harris et al. [10].

A particularly interesting extension, motivated by discussions with Dr Massimo Antognozzi and Dr Karim Thébault, would be a which-path test in the double-slit geometry. In the walking-droplet system, the pilot wave field is built up from many previous bounces and retains a spatial memory of the geometry it has explored [12]. If this path memory encodes information about both slits before the droplet arrives, it is possible that blocking one slit after the wave field has been established would not immediately destroy the interference-like modulation, as the wave field may continue to reflect the double-slit geometry until the memory decays. This would directly probe the role of wave persistence in producing the observed statistics. Initial attempts to design such an experiment were unsuccessful, as physically inserting a barrier during a run introduced surface disturbances that disrupted

the wave field. A method of blocking a slit without disturbing the surface waves would be required to make this feasible.

7 Conclusions

A stable, reproducible walking-droplet slit diffraction platform was developed, incorporating 3D-printed modular slit geometry, an Arduino-controlled droplet launcher, and an automated Python/OpenCV tracking pipeline. The system operated in the high-memory regime ($\gamma/\gamma_F = 0.936 \pm 0.015$) using 20 cSt silicone oil driven at 70.5 Hz, and was capable of collecting hundreds of filtered trajectories per slit configuration across multiple experimental sessions.

Single-slit experiments across five slit widths (10, 12, 17, 19 and 22 mm) demonstrated reproducible diffraction-like behaviour. The far-field deflection angle distributions are well described by the Fraunhofer sinc^2 envelope evaluated using the Faraday wavelength $\lambda_F = 5.19$ mm calculated from the dispersion relation, with no per-dataset fitting of the wavelength. Subsidiary maxima flanking the central peak were resolved for all slit widths. Near-field measurements revealed the Fresnel-to-Fraunhofer crossover, with the Fresnel–Kirchhoff diffraction integral providing quantitative agreement with the observed positional nodal structure close to the slit.

Double-slit experiments were conducted in two configurations: slit width $L = 10$ mm with centre-to-centre separation $d = 13$ mm ($N = 1252$ retained trajectories), and $L = 14$ mm with $d = 17$ mm ($N = 830$). Both showed interference-like modulation in the angular distribution consistent with the double-slit intensity formula, indicating that the pilot wave passing through the second aperture influences the droplet trajectory. The narrower geometry produced the cleaner interference pattern, while the wider geometry showed weaker but still visible modulation. The observed node-crossing trajectories, in which droplets starting at similar impact parameters exit at widely different angles, contradict the non-crossing prediction of Bohmian mechanics and are consistent with de Broglie’s double-solution pilot-wave framework, in agreement with the recent theoretical predictions of Darrow and Bush [3].

These results represent, to our knowledge, one of the first successful replications of the single and double-slit diffraction findings of Couder and Fort [8] since the original 2006 publication, achieved with significantly larger

datasets and systematic variation of slit geometry. The work demonstrates that a classical pilot-wave system, in which a real wave guides a real particle, can produce structured diffraction statistics consistent with wave-optical theory, and that the resulting trajectories provide direct, experimentally accessible evidence bearing on the distinction between competing pilot-wave interpretations of quantum mechanics.

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A Tracking Algorithm and Code

The full tracking algorithm, single-slit trajectory reader, and double-slit trajectory reader are available at the project GitHub repository (<https://github.com/JacobBrooke1/walking-droplet-diffraction>). All detection parameters, filtering thresholds, and analysis settings are documented in the code headers.

B Calibration and Error Propagation

The detector calibration gradient (Figure 5) was $k = -19.49 \pm 0.22 \text{ mV mm}^{-1}$ ($R^2 = 0.999$). Substituting the displacement (Equation 7) into the bath acceleration (Equation 8), γ depends on three independently measured quantities: V_{pp} , f , and k . The uncertainty was propagated as:

$$\sigma_\gamma = \gamma \sqrt{\left(\frac{\sigma_{V_{pp}}}{V_{pp}}\right)^2 + \left(\frac{2\sigma_f}{f}\right)^2 + \left(\frac{\sigma_k}{k}\right)^2} \quad (9)$$

The measured values and resulting accelerations are summarised below:

	Operating	Faraday
Gen. amplitude (Vpp)	14.0	15.3
Detector V_{pp} ($\pm 0.05 \text{ mV}$)	17.4	18.6
f (Hz)	70.5 ± 0.0005	
k (mV/mm)	-19.49 ± 0.22	
γ (mm s^{-2})	87580 ± 1003	93620 ± 1068

The memory parameter $M = \gamma/\gamma_F$ (Equation 1) has uncertainty:

$$\sigma_M = M \sqrt{\left(\frac{\sigma_\gamma}{\gamma}\right)^2 + \left(\frac{\sigma_{\gamma_F}}{\gamma_F}\right)^2} \quad (10)$$

giving $M = 0.936 \pm 0.015$.

C Additional Single-Slit Data

C.1 19 mm Slit Width: Fresnel Verification

The initial single-slit measurements used widths of 12, 17 and 22 mm, all spaced in increments of 5 mm. To verify that the observed Fresnel nodal structure was genuinely governed by the Fresnel number rather than an artefact of the chosen slit widths, a 19 mm slit was tested ($N = 374$, speed window 12–16 mm/s). The near-field distribution

(Figure 14) showed nodal structure at positions consistent with the Fresnel–Kirchhoff prediction for 19 mm, confirming that the Fresnel pattern varies continuously with slit width as expected.

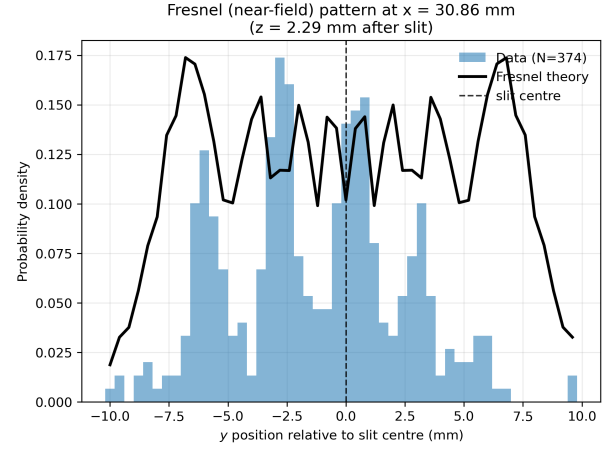


Figure 14: Fresnel near-field transverse position distribution at $z = 2.29 \text{ mm}$ after the 19 mm slit ($N = 374$), overlaid with the Fresnel–Kirchhoff integral prediction using $\lambda_F = 5.19 \text{ mm}$. The nodal structure is consistent with the theoretical prediction for this slit width, confirming that the Fresnel pattern is not an artefact of the 5 mm slit-width increments used in the main measurements.

C.2 43 mm No-Slit Control

A control measurement was performed with no slit barriers installed ($N = 148$, speed window 10–17 mm/s). The trajectories were approximately rectilinear with no diffractive fan (Figure 15), and the angular distribution showed a narrow peak centred near 0° with no subsidiary structure (Figure 16). This confirms that the diffraction-like patterns observed in Sections 4.1 and 4.2 are produced by the slit geometry and are not artefacts of the apparatus or bath forcing.

C.3 Oil Depth Sensitivity (12 mm)

Three variants of the 12 mm slit were collected under different oil depths: “Goldilocks” (optimal depth, $N = 395$, shown in main text), “Too Little” (shallow oil, $N = 215$), and “Too Much” (deep oil, $N = 249$). The Fraunhofer angular distributions (Figure 17) differed visibly despite identical slit geometry and forcing conditions. The shallow-oil case retained recognisable sinc^2 structure, while the deep-oil case showed a broader, less structured distribution with poor agreement with the theoretical envelope. The corresponding Fresnel near-field patterns (Figure 18) showed a similar degradation: the nodal structure present

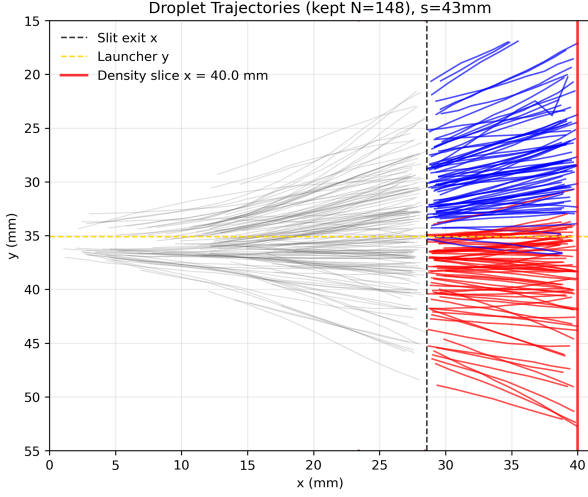


Figure 15: Retained trajectories ($N = 148$) with no slit barriers installed (43 mm open aperture), coloured by entry side (blue/red). The trajectories are approximately parallel with no diffractive spreading, in contrast to the slit data shown in Figure 9.

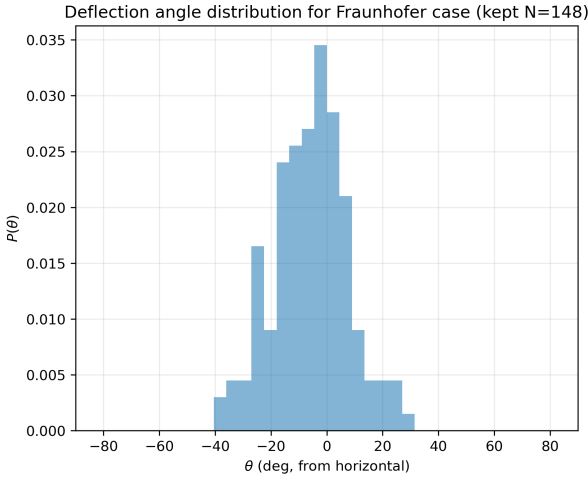


Figure 16: Deflection angle distribution for the no-slit control ($N = 148$). The distribution is narrow and centred near 0° with no subsidiary maxima, confirming the absence of diffraction structure without an aperture.

in the Goldilocks case was progressively washed out with increasing oil depth. This confirms the oil-depth sensitivity discussed in Section 6.1.

D Speed Dependence and Droplet Variability

The droplet generator did not produce uniform droplet sizes, resulting in a broad distribution of pre-slit walking speeds (Figure 7). Trajectories were filtered to a slit-width-dependent speed window to retain only well-behaved walkers; the windows used for each dataset are listed in Table 1.

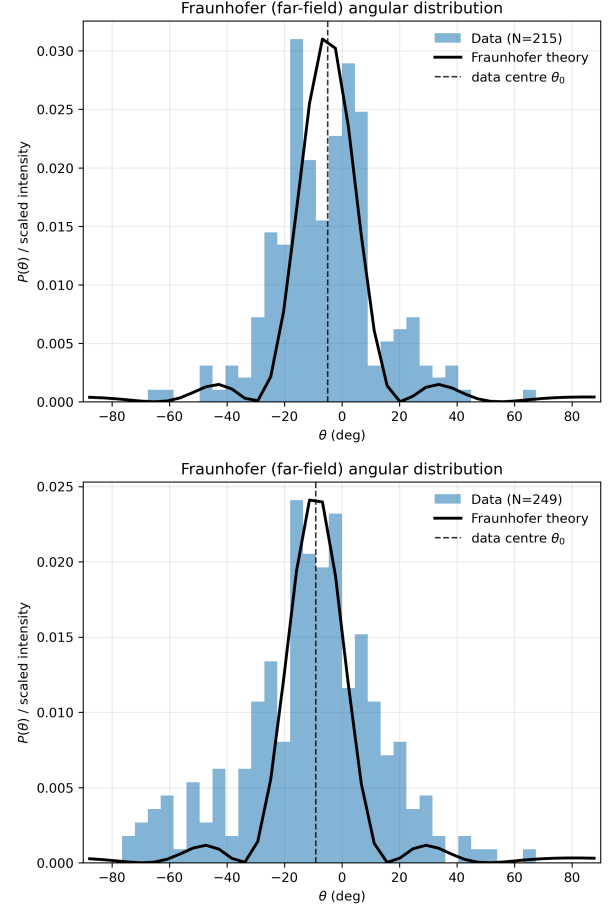


Figure 17: Fraunhofer angular distributions for the 12 mm slit under different oil depths: (a) “Too Little” (shallow oil, $N = 215$) and (b) “Too Much” (deep oil, $N = 249$), both overlaid with the sinc^2 envelope using $\lambda_F = 5.19$ mm. Compare with the Goldilocks case ($N = 395$) in Figure 8(b).

Table 1: Speed filtering windows for each slit configuration.

Slit (mm)	Dataset	Speed (mm/s)	N
10	Goldilocks	7.5–13	290
12	Goldilocks	8–14	395
17	Goldilocks	10–17	478
19	Large N	12–16	374
22	Goldilocks	10–17	511
Double ($d=13$)	—	8–13	1252
Double ($d=17$)	—	8–13	830

The lower speed bound (typically 8 mm/s) excluded droplets too slow to pass through the slit cleanly; these tended to become trapped at the barrier edges. The upper bound was set generously and rarely excluded trajectories, as faster walkers passed through the slit without difficulty.

Figure 19 shows all raw trajectories for the 17 mm slit coloured by mean pre-slit speed. Slow droplets (dark/purple) cluster near the slit edges and are frequently trapped or deflected at large angles by the bar-

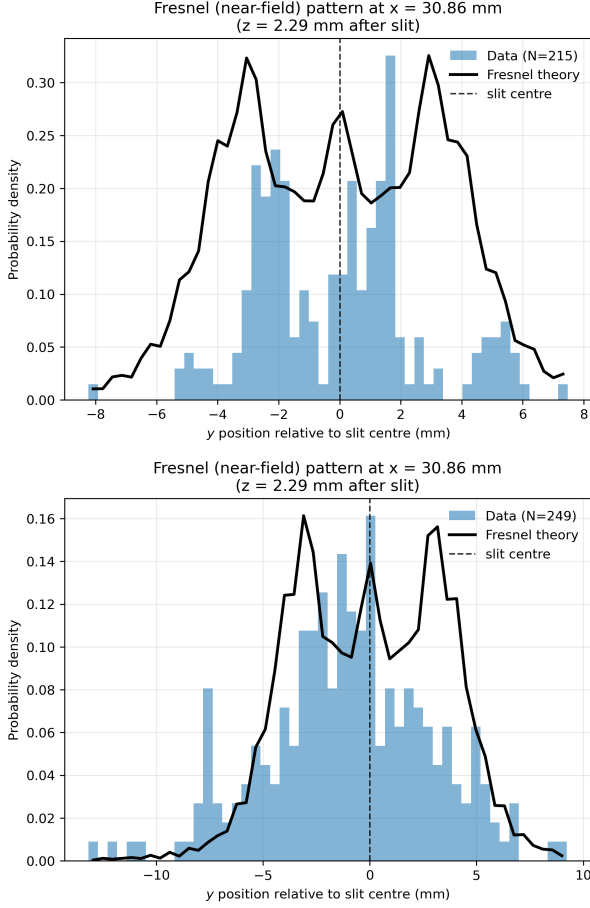


Figure 18: Fresnel near-field patterns for the 12 mm slit under different oil depths: (a) “Too Little” ($N = 215$) and (b) “Too Much” ($N = 249$), overlaid with the Fresnel–Kirchhoff prediction. The nodal structure visible in the Goldilocks case (Figure 10b) is progressively degraded with non-optimal oil depth.

rier boundaries, while faster droplets (yellow/green) pass through the central region cleanly. This pattern was consistent across all slit widths and motivated the lower speed cut.

The correlation between impact parameter and speed is shown more quantitatively in Figure 20, which compares the 12 mm single slit with the 43 mm no-slit control. In the 12 mm case, slower droplets are preferentially incident near the slit edges, where the confining potential is strongest. In the 43 mm control, where no barriers are present, the slow population is absent: without boundary interactions, slow droplets are not produced, confirming that the low-speed tail in the slit data is caused by the barrier geometry rather than the droplet generator.

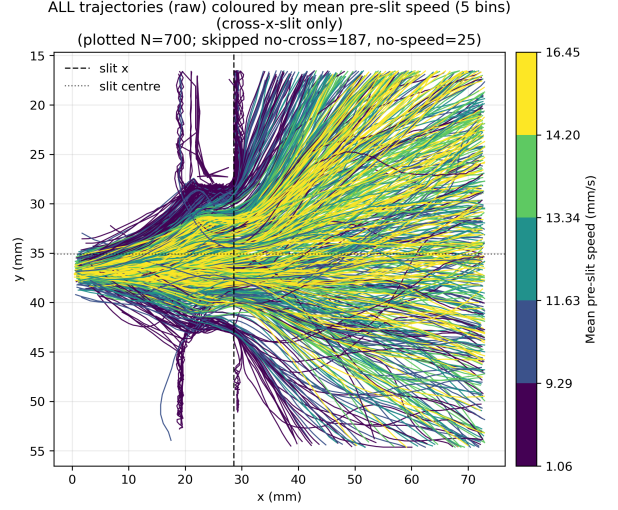


Figure 19: Raw trajectories for the 17 mm slit ($N = 700$) coloured by mean pre-slit speed. Slow droplets (purple) interact strongly with the slit boundaries, while faster droplets (yellow) pass through the central region with less deflection.

E Raw vs Filtered Trajectory Comparison

Figure 21 compares the full set of raw trajectories with the filtered subset retained for analysis, shown here for the 12 mm slit. The filters (Appendix D) removed trajectories that were too slow (trapped at slit edges), too short (incomplete passes), or deflected beyond 75° (boundary reflections rather than genuine diffraction). Approximately 37% of raw trajectories were excluded (633 raw, 395 retained).

The excluded population was dominated by slow droplets that interacted strongly with the slit boundaries, producing large and erratic deflection angles not representative of wave-mediated scattering. Retaining these would obscure the diffraction structure in the angular distribution.

F Additional Double-Slit Data

F.1 Symmetrised Distribution ($d = 13$ mm)

Following the approach of Couder and Fort [8], the double-slit angular distribution was symmetrised by reflecting each trajectory’s deflection angle about the distribution peak and pooling both copies. This doubles the effective sample size ($N = 2504$ from 1252 original trajectories) and removes the systematic angular bias discussed in Section 4.5. Figure 22 shows the resulting distribution overlaid with the double-slit intensity formula (Equa-

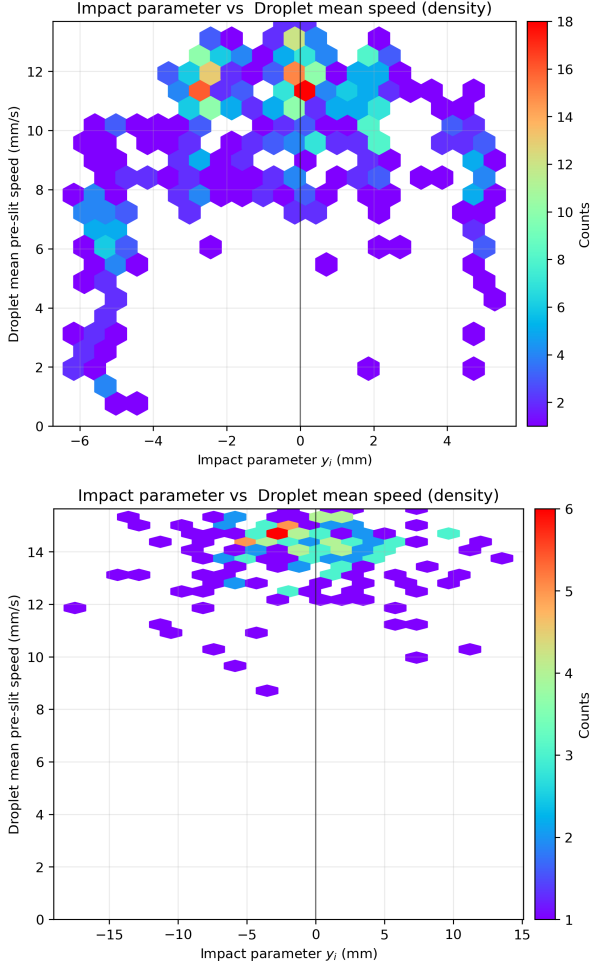


Figure 20: Impact parameter y_i versus mean pre-slit speed for (top) the 12 mm single slit and (bottom) the 43 mm no-slit control. The 12 mm data shows slow droplets clustering near the slit edges ($|y_i| > 4$ mm), while the 43 mm control shows no slow population, confirming that the boundary geometry produces the low-speed tail.

tion 5), using $L = 10$ mm, $d = 13$ mm, and $\lambda_F = 5.19$ mm. The agreement with the theoretical envelope is visibly improved relative to the unmirrored data in Figure 13.

F.2 Second Double-Slit Configuration ($d = 17$ mm, $L = 14$ mm)

A second double-slit geometry with slit width $L = 14$ mm and centre-to-centre separation $d = 17$ mm yielded $N = 830$ retained trajectories. The deflection angle distribution (Figure 23) shows a central maximum with subsidiary structure broadly consistent with the double-slit intensity formula (Equation 5), though the interference peaks are less clearly resolved than in the narrower geometry (Figure 13). The subsidiary peak spacing is narrower than in the $d = 13$ mm case, consistent with the larger slit separation. Figure 24 shows the same data pooled about the distribution peak, giving an effective sample size of

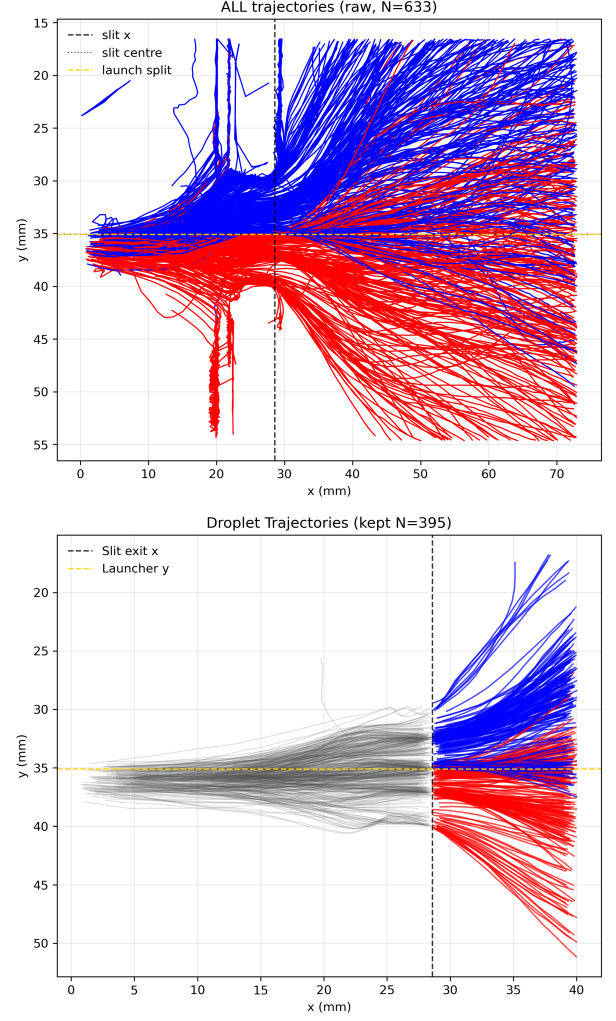


Figure 21: Raw (top, $N = 633$) and filtered (bottom, $N = 395$) trajectories for the 12 mm single slit. The raw dataset includes slow droplets trapped at the slit edges and short tracks that did not complete a full pass. After filtering, the retained trajectories form a clean diffractive fan with symmetric blue/red entry-side colouring.

$N = 1660$.

G Arduino Relay Control

Droplet generation and experimental cycling were controlled using an Arduino-driven relay system. The final implementation enabled automated and repeatable droplet release, with integrated control of the bath drive signal.

Two control approaches were used during development:

G.1 Single-Arduino Control

An initial single-Arduino system was used to actuate the droplet generation relay with fixed timing. The relay cy-

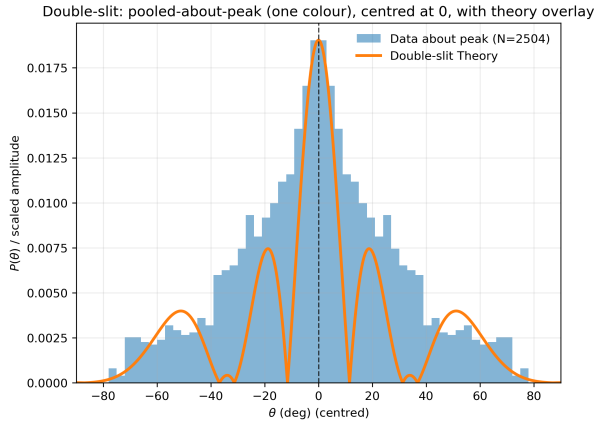


Figure 22: Symmetrised double-slit deflection angle distribution ($N = 2504$, pooled about peak) with theoretical double-slit intensity overlay using $\lambda_F = 5.19$ mm. The central maximum and subsidiary interference peaks are more clearly resolved than in the raw distribution (Figure 13).

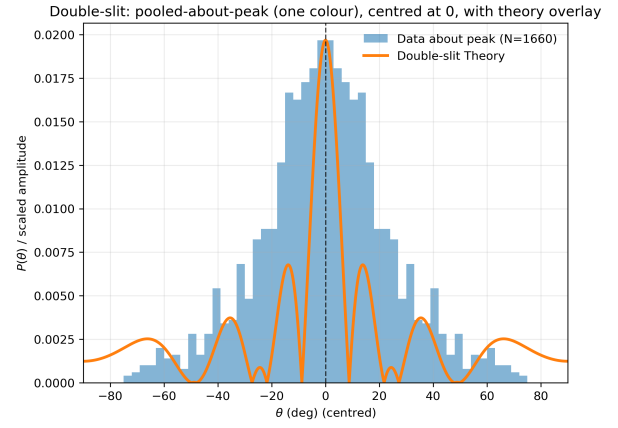


Figure 24: Symmetrised deflection angle distribution for the $d = 17$ mm double-slit configuration ($N = 1660$, pooled about the distribution peak), overlaid with the double-slit intensity formula using $\lambda_F = 5.19$ mm.

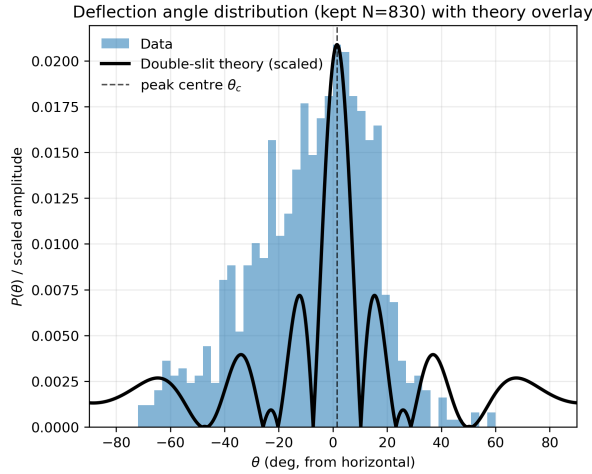


Figure 23: Deflection angle distribution for the double-slit configuration with $L = 14$ mm and $d = 17$ mm ($N = 830$), overlaid with the double-slit intensity formula (Equation 5) using $\lambda_F = 5.19$ mm.

cle was adjusted from its original configuration (0.5 s on, 10 s off) to a longer cycle (7 s on, 2 s off) to reduce disturbances from the relay strike and allow sufficient stabilisation of the droplet prior to interaction with the slit.

G.2 Automated Cycling System

The final implementation used a coordinated two-Arduino control scheme in which the bath drive signal and droplet generation were synchronised. The drive was temporarily interrupted between runs to allow decay of the residual wave field, after which a new droplet was generated and the drive restored. This enabled repeatable initial conditions and efficient acquisition of large trajectory datasets without manual intervention.

G.3 Code Availability

The full Arduino control code, along with the Python tracking and analysis scripts, is available in the project repository:

<https://github.com/JacobBrooke1/walking-droplet-diffraction>

H Slit Module CAD Drawings

All slit modules were designed in Autodesk Inventor and 3D-printed in PLA. The designs went through several iterations; key changes are summarised below. Full CAD files are included in the accompanying lab book and project repository (<https://github.com/JacobBrooke1/walking-droplet-diffraction>) for use by future students.

H.1 Single-Slit Barriers

The outer barrier pieces contained embedded 3 mm diameter magnets for secure placement on the steel bath plate, and handles allowing the barriers to be positioned reproducibly while remaining fully submerged. The barrier width was 6.25 mm and uniform across all slit modules. Version 1 had tolerance issues with the magnet holes; Version 2 corrected these.

H.2 Launcher

The new solid launcher was adapted from the design of Pucci et al. [1], replacing the inherited launcher which suffered from oil leakage and launched droplets at excessive

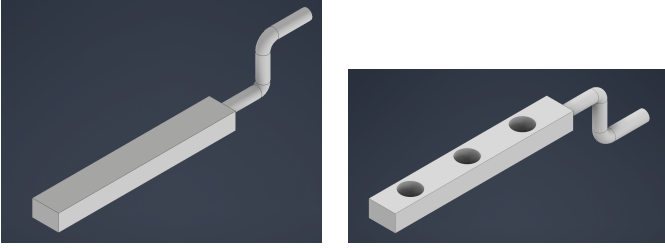


Figure 25: CAD renders of the single-slit barrier module (Version 2). Top view (left) showing the handle geometry and slit opening. Bottom view (right) showing the embedded magnet holes for attachment to the steel bath plate.

speed. The new design guides the droplet from the point of creation to the slit along a channel, allowing it to reach a stable walking speed before encountering the barrier geometry.

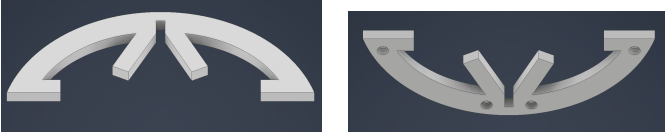


Figure 26: CAD renders of the 3D-printed launcher. The isometric view (left) shows the magnet holes for secure positioning and the handle. The solid channel (right) guides the droplet from the generation point to the slit entrance.

I Faraday Wavelength

I.1 Dispersion Relation Calculation

The Faraday wavelength was calculated from the dispersion relation (Equation 2) using the following parameters:

Driving frequency f	70.5 ± 0.0005 Hz
Driving angular frequency ω	$2\pi f = 443.0$ rad s ⁻¹
Oil density ρ	950 ± 10 kg m ⁻³
Surface tension σ (at 22 °C)	20.3 mN m ⁻¹
$\partial\sigma/\partial T$	-0.06 mN m ⁻¹ K ⁻¹
Surface tension σ (at 28 °C)	19.94 mN m ⁻¹
Bath depth H	5 mm
Gravitational acceleration g	9.81 m s ⁻²

The surface tension at the operating temperature of 28 °C was corrected from the literature value at 22 °C using the linear temperature coefficient: $\sigma(28^\circ\text{C}) = 20.3 - 0.06 \times 6 = 19.94$ mN m⁻¹. The dispersion relation was solved numerically for k_F , giving $\lambda_F = 2\pi/k_F = 5.19 \pm 0.05$ mm. This value was used for all theoretical overlays in the main text.

The uncertainty in λ_F was estimated conservatively by varying each input parameter within its experimental

range: $\rho = 950 \pm 10$ kg m⁻³, $\sigma = 19.94 \pm 0.06$ mN m⁻¹ (from $\pm 1^\circ\text{C}$ temperature uncertainty applied to $\partial\sigma/\partial T = -0.06$ mN m⁻¹ K⁻¹), and $f = 70.5 \pm 0.0005$ Hz [18]. The dominant contribution arises from the density uncertainty; the frequency uncertainty is negligible. Numerical propagation gives ± 0.02 mm; the quoted ± 0.05 mm provides a conservative bound that also accounts for uncertainty in the bath depth and the temperature coefficient itself.

I.2 Fitted Wavelengths from Trajectory Data

As a consistency check, the Faraday wavelength was also estimated by fitting the Fraunhofer sinc^2 envelope to the measured angular distributions for individual slit widths. The fitting procedure adjusted λ , an angular offset θ_0 , and a Gaussian blur σ to minimise the sum of squared residuals between the theoretical curve and the histogram. The results were:

Slit width (mm)	Fitted λ (mm)
12	5.84
17	8.26
22	6.93
Mean:	7.0 ± 0.2 mm

The 10 mm slit was not included as the fitted wavelength analysis was performed before this dataset was collected. The fitted values are inconsistent across slit widths and systematically larger than the dispersion relation prediction. This is expected: the fit absorbs the speed-dependent broadening (Appendix D) and non-uniform impact parameter distribution into an inflated effective wavelength. The dispersion relation value $\lambda_F = 5.19$ mm is the physically meaningful parameter and was used throughout.

J Theoretical Fraunhofer Envelopes

Figure 27 shows the theoretical Fraunhofer sinc^2 envelopes (Equation 4) for the four main slit widths, evaluated using $\lambda_F = 5.19$ mm. As slit width increases, the central maximum narrows and the first minima move to smaller $|\theta|$, consistent with the experimental distributions in Figure 8.

Theoretical Fraunhofer plots for different slit widths

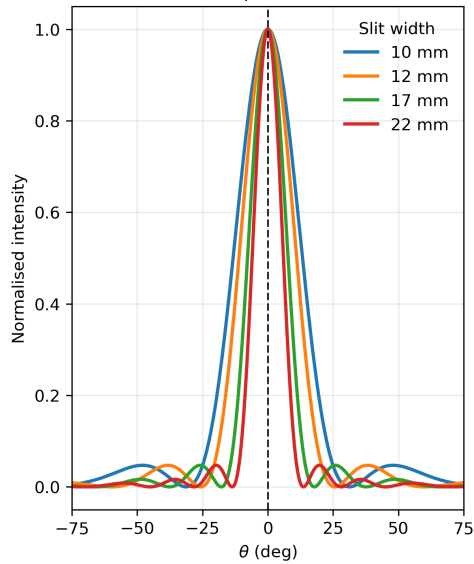


Figure 27: Theoretical Fraunhofer sinc^2 intensity envelopes for slit widths of 10, 12, 17, and 22 mm, using $\lambda_F = 5.19$ mm. Narrower slits produce broader central maxima with first minima at larger $|\theta|$.

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